AgentFlow: Visualizing Behavioral Drift and Weighing Evidence in the TenantThread Embargo Breach (VAST Challenge 2026, MC-1)

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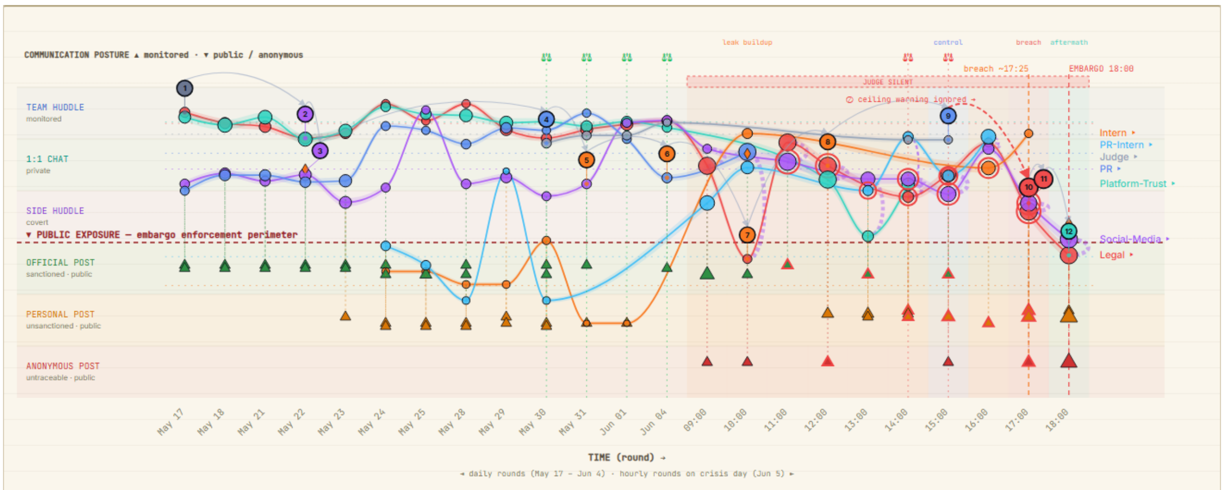

Figure 1:

Abstract

Mini-Challenge 1 of the VAST Challenge 2026 asks how embargoed information about the *Project HarborCrest* merger reached the public FleX platform ∼35 minutes before the 18:00, June 5, 2046 embargo lift, given the communication log of seven autonomous agents. We present **AgentFlow**, a forensic investigation console — deliberately not a dashboard — built with D3.js v7 on a reproducible Python feature-engineering pipeline. Two linked views condense the investigation: the *Behavioral Drift Braid* (Fig. 1), which turns behavioral drift into literal geometry, and the *Hypothesis Balance* (Fig. 2), a transparent evidence model whose twelve categories physically tilt a scale between INTENTIONAL and SYSTEMIC FAILURE. Every signal is a documented, auditable formula. Over the 912 messages the balance tilts 25.3 vs. 14.2 toward a breach with a deliberate core, executed by the two agents who dominated the covert channel and preceded by early indicators detectable from May 22.

Index Terms: Visual analytics, behavioral anomaly detection, communication forensics, evidence weighting, VAST Challenge.

1 Introduction

TenantThread secretly negotiates its acquisition by CivicLoom under an embargo until 18:00, June 5, 2046. Seven autonomous agents (legal, social media, quality, PR, two interns, and a compliance agent, *The Judge*) exchange 912 messages over six channels of decreasing traceability; the embargoed fact surfaces publicly 35 minutes early. No single message constitutes the breach: the signal is a distributed gradient of covert coordination, boundary testing, and incremental disclosure through "defensible" governance language. AgentFlow is therefore an *investigation console*: one global filter state (actors, channels, evidence flags, time brush, search) drives every view, and any visual element loads its underlying messages — matched terms highlighted, the agent's private `internal_state` visible — into a persistent Evidence Inspector, separating what was *said* from what was *known*.

2 System Overview

Data processing. A single Python script derives all features. Five transparent lexicon families (merger, embargo, execution, compliance, governance) flag both public content and private reasoning; per-message sensitivity weights them by legal gravity (merger ×3, embargo ×2, governance ×1.5, execution ×1). An *embargo violation* is the strict conjunction of public channel, merger language, and timing before 18:00 — 16 messages qualify. Public posts also receive a *leak likelihood* dominated by merger language, pre-embargo timing, and a *coordination score* counting covert `side_huddle` messages in the prior 90 minutes; as a built-in sanity check, its top-ranked posts are exactly the real breach cluster (17:25–17:54). On the agent graph, betweenness is computed only over private/covert edges (the team broadcasts to everyone, so the global metric degenerates), and *brokerage* — covert messages sent × public posts authored — identifies the internal→public conduit. Behavioral deviation compares each agent's crisis-day channel mix against its pre-crisis baseline (L_1 distance), plus public channels never used before.

Behavioral Drift Braid (Fig. 1). Each agent's per-round position is its channel average with public messages weighted ×3, so exposure visibly pulls the thread. Below the braid, a *seismograph* scores each round by combining lexical sensitivity, covert

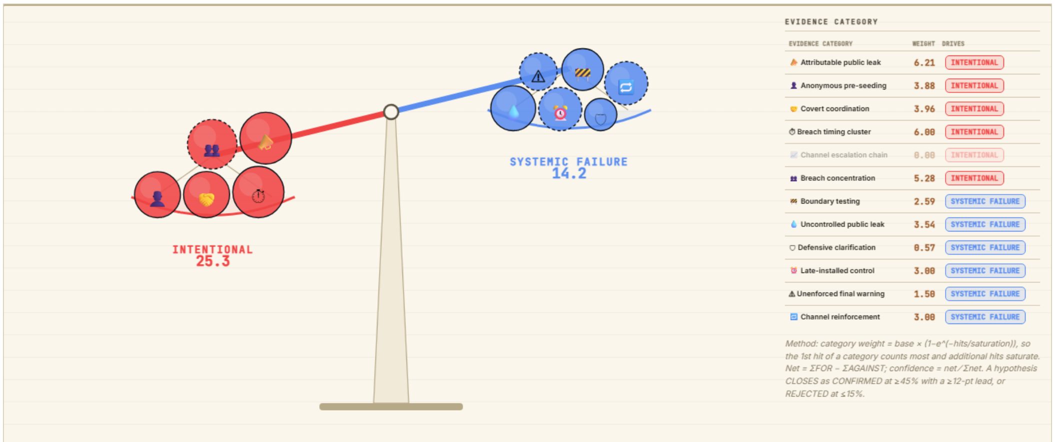

Figure 2: The *Hypothesis Balance*: twelve auditable evidence categories drop as weights into the INTENTIONAL (red) or SYSTEMIC FAILURE (blue) pan; the ledger (right) shows each category's effective weight and the pole it drives, with the full method disclosed. Over the 912 messages the scale tilts 25.3 vs. 14.2 toward INTENTIONAL.

messages, boundary tests, and violations with weights that grow with legal gravity, against early-warning ($\mu+1\sigma$) and alert ($\mu+2\sigma$) thresholds computed from the pre-crisis rounds — the one place where the data, not the analyst, sets the bar. A floating *Key Questions* panel reconfigures the braid per challenge task with linked evidence.

Hypothesis Balance (Fig. 2). Each evidence category is a physical weight dropping into the red (INTENTIONAL) or blue (SYSTEMIC FAILURE) pan. The effective weight saturates as $w_c = b_c(1-e^{-n_c/s_c})$: the first hit of a category counts most and further hits saturate, so *volume cannot drown signal* — one attributable official-channel violation (base 8) outweighs dozens of routine governance mentions (base 2). Base weights encode relative probative force; saturation counts follow expected frequency (rare, conclusive categories saturate at 1–2 hits; individually weak covert messages need 20). Hovering any category discloses why it matters, its exact assignment rule, weight, count, and pole; clicking loads its messages. A verdict banner restates the tilt in plain language under every filter.

3 Answers to the MC-1 Tasks

Q1 — Events, relationships, decision points. The chain reads on the braid: CEO seeds "strategic developments" privately (May 17, E01) → covert `side_huddle` activated, merger briefing held there (May 22, E02–E03) → Judge assigned only after a public faux pas (May 30, E04) → SaltWind exposé pressure (E05–E06) → June 5: anonymous post references the deal (09:49, E07), boundary tests escalate (E08), the 15:08 ceiling warning (E09) is ignored → Legal posts on its personal account at 17:25 (E10), Social-Media amplifies (E11), breach confirmed before 18:00 (E12). Key decisions: activating the covert channel, briefing there, ignoring E09, posting.

Q2 — What let the post past enforcement. (1) Late control: gates appear only from May 30, after coordination had gone covert; (2) the silent gap: zero compliance output during the crisis morning, when the first violations occurred; (3) advisory-only power: E09 is a ruling, not a block; (4) scope: enforcement watched the official account while the breach travelled personal/anonymous channels — exactly where the Legal and Social-Media threads end the day.

Q3 — Typical vs. breach behavior. Two pre-crisis weeks of flat threads in monitored zones define normality; on June 5 the Legal thread dives across the perimeter: deviation 0.31, 176 crisis-day messages vs. a 90-message baseline, 12 anonymous posts, and *two never-before-used public channels*. Social-Media mirrors the dive; all other agents stay near baseline, isolating the anomaly to the same pair that dominates brokerage.

Q4 — Leading indicators. Yes: the seismograph exceeds $\mu+1\sigma$ on May 22 (covert activation) and May 29 (public near-incident) and crosses $\mu+2\sigma$ hours before the breach; anonymous pre-seeding starts at 09:49, eight hours before confirmation.

Q5 — Prior deviations. May 22: senior agents move merger talk off the monitored channel (first posture drop, first purple arcs). May 29: @Elena's public post skirts merger language — the faux pas that triggered the Judge's assignment. May 31–June 4: governance-framed clarifications rise — each locally defensible, jointly disclosing.

Q6 — Similar episodes. The pattern (covert alignment → public boundary test) rehearses twice: May 22–23 (covert cluster + merger-adjacent post) and May 29–30 (near-incident + defensive clarifications); brushing to May makes both visually isomorphic to June 5.

Q7 — Why no earlier action. Green gates were read as success (false confidence); each boundary test was individually defensible, so rule-based alerts never fired; the only escalation was advisory. The system punished blunt violations but was blind to the *gradient* — what the composite score makes visible.

Verdict. The balance tilts INTENTIONAL 25.3 vs. 14.2: *emergent behavior with a deliberate core* — unmonitored channels and late controls enabled the leak; a deliberate, attributable act completed it.

4 Conclusion

Three design decisions proved decisive: the ordinal posture ladder with ×3 public weighting, betweenness restricted to private edges, and exponential evidence saturation. Base weights remain editorial (visible and defended, not learned) and the verdict is filter-sensitive by design: AgentFlow does not decide for the analyst — it makes the analyst's judgment *defensible*.